

## §2 量子力学的形式理论

### §2.1 左/右矢, Hilbert 空间

1. 态叠加原理: 任何一个量子态是其它几个量子态的叠加,  
几个量子态的叠加产生新的量子态.

(1)  $\hookrightarrow$  量子态  $\sim |\alpha\rangle$  ket

$$\textcircled{1} |\alpha\rangle + |\beta\rangle = |\gamma\rangle$$

$$\textcircled{2} |\alpha\rangle + |\beta\rangle = |\beta\rangle + |\alpha\rangle,$$

$$(|\alpha\rangle + |\beta\rangle) + |\gamma\rangle = |\alpha\rangle + (|\beta\rangle + |\gamma\rangle)$$

$$\textcircled{3} \exists |\text{null}\rangle \text{ s.t. } |\alpha\rangle + |\text{null}\rangle = |\alpha\rangle,$$

$$, \forall |\alpha\rangle, \exists |\alpha'\rangle \text{ s.t. } |\alpha\rangle + |\alpha'\rangle = |\text{null}\rangle$$

$$\textcircled{4} c|\alpha\rangle = |\alpha\rangle c, \quad c|\alpha\rangle \text{ 与 } |\alpha\rangle \text{ 相同.}$$

$$\textcircled{5} (cd)|\alpha\rangle = c(d|\alpha\rangle)$$

$$(c+d)|\alpha\rangle = c|\alpha\rangle + d|\alpha\rangle$$

$$c(|\alpha\rangle + |\beta\rangle) = c|\alpha\rangle + c|\beta\rangle$$

$$1|\alpha\rangle = |\alpha\rangle$$

(2) 对偶空间  $\sim \langle\alpha|$  bra

$$-- \text{对 } \lambda_2: |\alpha\rangle \xrightarrow{\text{DC}} \langle\alpha|$$

$$c_\alpha |\alpha\rangle + c_\beta |\beta\rangle \xrightarrow{\text{DC}} c_\alpha^* \langle\alpha| + c_\beta^* \langle\beta|$$

$$\text{内积: } \langle\beta|\alpha\rangle = \langle\beta| \cdot |\alpha\rangle$$

$$\textcircled{1} \langle\beta|\alpha\rangle = \langle\alpha|\beta\rangle^*$$

$$\textcircled{2} \langle\alpha|\alpha\rangle \geq 0$$

$$\textcircled{3} \langle\gamma|(c_\alpha |\alpha\rangle + c_\beta |\beta\rangle) = c_\alpha \langle\gamma|\alpha\rangle + c_\beta \langle\gamma|\beta\rangle$$

$$\text{模: } \|\alpha\| = \sqrt{\langle\alpha|\alpha\rangle}$$

$$\text{正交: } \langle\alpha|\beta\rangle = 0$$

完备性: 线性空间中包含系统中所有的量子态.

完备的线性空间是 Hilbert 空间

## § 2.2 线性算符和可观测算.

1. 算符: 对态变量的变换

$$\text{线性: } \hat{A}(|\alpha\rangle + |\beta\rangle) = \hat{A}|\alpha\rangle + \hat{A}|\beta\rangle \quad |(\hat{A} + \hat{B}) + \hat{C} = \hat{A} + (\hat{B} + \hat{C})$$

$$\hat{A}(c|\alpha\rangle) = c\hat{A}|\alpha\rangle$$

$$(c\hat{A})|\alpha\rangle = c(\hat{A}|\alpha\rangle)$$

$$\text{相等: } \hat{A} = \hat{B} \Leftrightarrow \hat{A}|\alpha\rangle = \hat{B}|\alpha\rangle \quad \forall |\alpha\rangle$$

$$\text{厄米共轭: } \hat{A}^\dagger := |\alpha\rangle \xrightarrow{\hat{A}} \langle\alpha| \hat{A}^\dagger$$

$$(\hat{A}|\alpha\rangle)^\dagger = |\alpha\rangle^\dagger \hat{A}^\dagger = \langle\alpha| \hat{A}^\dagger \quad \underline{(|\alpha\rangle^\dagger = \langle\alpha|)}$$

$$(c_\alpha|\alpha\rangle + c_\beta|\beta\rangle)^\dagger = c_\alpha^* \langle\alpha| + c_\beta^* \langle\beta|$$

$$(c_\alpha \hat{A} + c_\beta \hat{B})^\dagger = c_\alpha^* \hat{A}^\dagger + c_\beta^* \hat{B}^\dagger$$

$$\hat{A}^\dagger = \hat{A} \rightarrow \hat{A} \text{ 是厄米算符.}$$

$$\text{乘法: } (\hat{A}\hat{B})|\alpha\rangle = \hat{A}(\hat{B}|\alpha\rangle)$$

$$\hat{A}\hat{B} \neq \hat{B}\hat{A}$$

$$\hat{A}(\hat{B}|\alpha\rangle) = (\hat{A}\hat{B})|\alpha\rangle$$

$$(\hat{A}\hat{B})^\dagger = \hat{B}^\dagger \hat{A}^\dagger$$

$$\text{证: } \langle\alpha|(\hat{A}\hat{B})^\dagger = [(\hat{A}\hat{B})|\alpha\rangle]^\dagger = [\hat{A}(\hat{B}|\alpha\rangle)]^\dagger = (\hat{B}|\alpha\rangle)^\dagger \hat{A}^\dagger = \langle\alpha|\hat{B}^\dagger \hat{A}^\dagger$$

$$\text{对易子: } [\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A} \quad [\hat{A}, \hat{B}] = 0 \rightarrow [\hat{A}, \hat{B}] \text{ 对易, } \hat{A}\hat{B} = \hat{B}\hat{A}$$

$$\text{结合律: } \langle\alpha|(\hat{A}|\beta\rangle) = (\langle\alpha|\hat{A})|\beta\rangle = \langle\alpha|\hat{A}|\beta\rangle$$

$$\hookrightarrow \langle\alpha|\hat{A}|\beta\rangle = \langle\beta|\hat{A}^\dagger|\alpha\rangle^*$$

$$\text{证: } \langle\alpha|\hat{A}|\beta\rangle = \langle\alpha|(\hat{A}|\beta\rangle) = [(\hat{A}|\beta\rangle)^\dagger |\alpha\rangle]^\dagger = \langle\beta|\hat{A}^\dagger|\alpha\rangle^*$$

$$\text{恒等算符: } \hat{I}|\alpha\rangle = |\alpha\rangle, \quad \langle\alpha|\hat{I} = \langle\alpha|$$

$$\text{逆算符: } \hat{A}|\alpha\rangle = |\beta\rangle, \quad \hat{A}^{-1}|\beta\rangle = |\alpha\rangle$$

$$\hookrightarrow \hat{A}^{-1}\hat{A} = \hat{I}$$

态变量的外积:  $|\alpha\rangle\langle\beta|$

$$(|\alpha\rangle\langle\beta|)^\dagger = |\beta\rangle\langle\alpha|$$

$$\text{证: } (|\alpha\rangle\langle\beta|)^\dagger = \langle\beta|\delta\rangle^* |\alpha\rangle = \langle\delta|\beta\rangle |\alpha\rangle$$

投影算符:  $\hat{P}_\alpha = |\alpha\rangle\langle\alpha|$   $\hat{P}_\alpha |\beta\rangle = |\alpha\rangle\langle\alpha|\beta\rangle$

$$\hat{P}^\dagger = \hat{P}$$

$$\text{证: } (|\alpha\rangle\langle\alpha|)^\dagger = |\alpha\rangle\langle\alpha|$$

$$\hat{P}^2 = \hat{P}$$

$$\text{证: } |\alpha\rangle\langle\alpha||\alpha\rangle\langle\alpha| = |\alpha\rangle\langle\alpha| \quad (|\alpha\rangle \text{ 归一化})$$

线性算符:  $\hat{A}(c_\alpha|\alpha\rangle + c_\beta|\beta\rangle) = c_\alpha^* \hat{A}|\alpha\rangle + c_\beta^* \hat{A}|\beta\rangle$

2. 厄米算符的本征值, 本征矢量.

$$\text{本征方程: } \hat{A}|\alpha\rangle = \alpha|\alpha\rangle$$

定理: 厄米算符的本征值必为实数

$$\langle\alpha|\hat{A}|\alpha\rangle = \alpha \underbrace{\langle\alpha|\alpha\rangle}_{=1} \quad \text{实数} \rightarrow \alpha \text{ 为实数}$$

定理: 厄米算符属于不同本征值的本征矢量正交.

$$\text{证: } \begin{cases} \hat{A}|\alpha\rangle = \alpha|\alpha\rangle \\ \hat{A}|\beta\rangle = \beta|\beta\rangle \end{cases} \quad (\hat{A}|\beta\rangle)^\dagger = \langle\beta|\hat{A}^\dagger = \langle\beta|\hat{A}$$

$$\hookrightarrow \langle\beta|\hat{A} = \langle\beta|\beta$$

$$\langle\beta|\hat{A}|\alpha\rangle - \langle\beta|\hat{A}|\alpha\rangle = 0 = (\beta - \alpha)\langle\beta|\alpha\rangle$$

$$\alpha \neq \beta \rightarrow \langle\beta|\alpha\rangle = 0$$

3. 算符函数

解析函数  $f(z)$  的展开为:

$$f(z) = \sum_n f_n z^n \rightarrow f(\hat{A}) = \sum_n f_n \hat{A}^n$$

$$f(\hat{A})|\alpha\rangle = f(\alpha)|\alpha\rangle$$

请、算符本征值的集合.

$f(z)$  良定义: 收敛半径  $\leq$  谱半径

### 3. 测量和可观测

假设:  $\hat{A}$  (系统量) 态  $|\alpha\rangle = \hat{A}$  的本征态,

$\hat{A}$  的本征态形成完备集

$\hat{A}$  为可观测,  $\hat{A}$  的观测值为实数  $\rightarrow \hat{A}$  是厄米算符.

$$|\alpha\rangle = \sum_n \phi_n |n\rangle, \quad \phi_n = \langle n | \alpha \rangle$$

测量后坍缩到  $|n\rangle$  的概率:  $P_n = |\langle n | \alpha \rangle|^2$

完备性条件:  $\sum_n |n\rangle \langle n| = 1$

期待值:  $\langle \hat{A} \rangle_\alpha = \langle \alpha | \hat{A} | \alpha \rangle$

$$\begin{aligned} &= \sum_{n, n'} \langle \alpha | n \rangle \langle n | \hat{A} | n' \rangle \langle n' | \alpha \rangle \quad \langle n | \hat{A} | n' \rangle = a' \langle n | n' \rangle = a' \delta_{nn'} \\ &= \sum_{n, n'} \langle \alpha | n \rangle \langle n | \hat{A} | n' \rangle \langle n' | \alpha \rangle = a_n \delta_{nn'} \\ &= \sum_n a_n |\langle n | \alpha \rangle|^2, \quad \text{其中 } P_n = |\langle n | \alpha \rangle|^2 \end{aligned}$$

共同本征态:  $|\alpha\rangle$  为  $\hat{A}, \hat{B}$  的共同本征态.

$$\hat{A} |\alpha\rangle = a |\alpha\rangle, \quad \hat{B} |\alpha\rangle = b |\alpha\rangle$$

$$\hookrightarrow \hat{A} \hat{B} |\alpha\rangle = b \hat{A} |\alpha\rangle = ba |\alpha\rangle = \hat{B} \hat{A} |\alpha\rangle$$

$\hat{A}, \hat{B}$  对易

定理: 若  $\hat{A}, \hat{B}$  对易, 则必有共同本征态.

证:  $|\alpha\rangle$  是  $\hat{A}$  的本征态,

$$\hat{A} (\hat{B} |\alpha\rangle) = \hat{B} (\hat{A} |\alpha\rangle) = a (\hat{B} |\alpha\rangle)$$

本征值  $a$  对应的子空间为  $\hat{B}$  的不变子空间. 记为  $W_a$

$\hookrightarrow$  在  $W_a$  中可找到  $\dim(W_a)$  个 (共同) 本征态

$$\text{取 } W \in W = (W_1, \dots, W_n)$$

$$\forall W \in W = W' = (W'_1, \dots, W'_n)$$

$$\begin{cases} \hat{A} |\alpha\rangle = a |\alpha\rangle \\ \hat{B} |\alpha\rangle = b |\alpha\rangle \end{cases}$$

$$\hat{A}, \hat{B} \text{ Hermitian}$$

$$\langle \alpha | \hat{B} = \langle \alpha | b, \quad \langle \alpha | \hat{A} = \langle \alpha | a$$

$$\langle \alpha | \hat{B} \hat{A} | \alpha \rangle = ba \langle \alpha | \alpha \rangle$$

$$\langle \alpha | \hat{A} \hat{B} | \alpha \rangle = ab \langle \alpha | \alpha \rangle$$

定理:  $[\hat{F}, \hat{A}] = 0$ ,  $[\hat{F}, \hat{B}] = 0$ ,  $[\hat{A}, \hat{B}] \neq 0$

则  $\hat{F}$  的本征值有简并

证: 充分性:

$\hat{F}|\psi\rangle = f|\psi\rangle$ , 则  $\hat{A}|\psi\rangle$ ,  $\hat{B}|\psi\rangle$  也是  $\hat{F}$  本征态,

可选取  $\psi$  使  $\hat{A}|\psi\rangle$ ,  $|\psi\rangle$  是同一个态, 此时为使  $\hat{B}|\psi\rangle$  与  $\hat{A}|\psi\rangle$  不同.

$\dim W > 1 \rightarrow$  有简并.

必要性:

设  $\hat{F}$  有两个不简并本征态  $|\psi_1\rangle$ ,  $|\psi_2\rangle$ , 对应本征值为  $f$

$$\hat{A} = a_1 |\psi_1\rangle\langle\psi_1| + a_2 |\psi_2\rangle\langle\psi_2| \quad (a_1 \neq a_2)$$

$$\hat{B} = b |\psi_1\rangle\langle\psi_1| + b^* |\psi_2\rangle\langle\psi_2| \quad (b \neq 0)$$

$$[\hat{F}, \hat{A}] = \hat{F}\hat{A} - \hat{A}\hat{F}$$

$$= \hat{F}(a_1 |\psi_1\rangle\langle\psi_1| + a_2 |\psi_2\rangle\langle\psi_2|) - (a_1 |\psi_1\rangle\langle\psi_1| + a_2 |\psi_2\rangle\langle\psi_2|)\hat{F}$$

$$= a_1 f |\psi_1\rangle\langle\psi_1| + a_2 f |\psi_2\rangle\langle\psi_2| - a_1 f |\psi_1\rangle\langle\psi_1| - a_2 f |\psi_2\rangle\langle\psi_2|$$

$$= 0$$

$$[\hat{F}, \hat{B}] = 0$$

$$[\hat{A}, \hat{B}] = a_1 b |\psi_1|^2 |\psi_1\rangle\langle\psi_1| + a_2 b^* |\psi_2|^2 |\psi_2\rangle\langle\psi_2|$$

$$+ a_1 b^* \langle\psi_2|\psi_1\rangle |\psi_1\rangle\langle\psi_2| + a_2 b \langle\psi_1|\psi_2\rangle |\psi_2\rangle\langle\psi_1|$$

$$- a_1 b |\psi_1|^2 |\psi_1\rangle\langle\psi_1| - a_2 b^* |\psi_2|^2 |\psi_2\rangle\langle\psi_2|$$

$$- b^* a_1 \langle\psi_2|\psi_1\rangle |\psi_2\rangle\langle\psi_1| - b a_2 \langle\psi_1|\psi_2\rangle |\psi_1\rangle\langle\psi_2|$$

$$= (a_1 b^* - a_2 b) \langle\psi_1|\psi_2\rangle (|\psi_1\rangle\langle\psi_2| - |\psi_2\rangle\langle\psi_1|)$$

$$\neq 0$$

#### 4. 不确定关系

$$\Delta \hat{A} = \hat{A} - \langle \hat{A} \rangle_\alpha, \quad \text{其中 } \langle \hat{A} \rangle_\alpha = \langle \alpha | \hat{A} | \alpha \rangle$$

$$\text{涨落: } \langle (\Delta \hat{A})^2 \rangle_\alpha = \langle \hat{A}^2 - 2\hat{A}\langle \hat{A} \rangle_\alpha + \langle \hat{A} \rangle_\alpha^2 \rangle_\alpha = \langle \hat{A}^2 \rangle_\alpha - \langle \hat{A} \rangle_\alpha^2$$

Schwarz 不等式:

$$\langle \alpha | \alpha \rangle \langle \beta | \beta \rangle \geq \langle \alpha | \beta \rangle^2$$

$$\text{证: } (\langle \alpha | + c^* \langle \beta |)(| \alpha \rangle + c | \beta \rangle) = |\alpha + c\beta|^2 \geq 0$$

$$\text{令 } c = \frac{-\langle \beta | \alpha \rangle}{\langle \beta | \beta \rangle}$$

$$\hookrightarrow \langle \alpha | \alpha \rangle - \frac{(\langle \beta | \alpha \rangle)^2}{\langle \beta | \beta \rangle} \geq 0, \quad \text{D.}$$

$$\hookrightarrow \langle (\Delta \hat{A})^2 \rangle_\alpha \langle (\Delta \hat{B})^2 \rangle_\alpha \geq |\langle \Delta \hat{A} \Delta \hat{B} \rangle_\alpha|^2$$

$$\Delta \hat{A} \Delta \hat{B} = \frac{1}{2} [\hat{A}, \hat{B}] + \frac{1}{2} \{\hat{A}, \hat{B}\}$$

其中  $\{\hat{A}, \hat{B}\}$  为反对易子,  $\{\hat{A}, \hat{B}\} := \hat{A}\hat{B} + \hat{B}\hat{A}$

$$* [\hat{A}, \hat{B}]^\dagger = (\hat{A}\hat{B} - \hat{B}\hat{A})^\dagger = \hat{B}\hat{A} - \hat{A}\hat{B} = -[\hat{A}, \hat{B}], \quad \text{纯实数}$$

$$\{\hat{A}, \hat{B}\}^\dagger = \{\hat{A}\hat{B} + \hat{B}\hat{A}\}^\dagger = \hat{B}\hat{A} + \hat{A}\hat{B} = \{\hat{A}, \hat{B}\}, \quad \text{纯虚数}$$

$$|\langle \Delta \hat{A} \Delta \hat{B} \rangle_\alpha|^2$$

$$= |\langle \frac{1}{2} [\Delta \hat{A}, \Delta \hat{B}] + \frac{1}{2} \{\Delta \hat{A}, \Delta \hat{B}\} \rangle_\alpha|^2$$

$$= \frac{1}{4} |\langle [\Delta \hat{A}, \Delta \hat{B}] \rangle_\alpha|^2 + \frac{1}{4} |\langle \{\Delta \hat{A}, \Delta \hat{B}\} \rangle_\alpha|^2$$

$$\hookrightarrow \langle (\Delta \hat{A})^2 \rangle_\alpha \langle (\Delta \hat{B})^2 \rangle_\alpha \geq \frac{1}{4} |\langle [\Delta \hat{A}, \Delta \hat{B}] \rangle_\alpha|^2 + \frac{1}{4} |\langle \{\Delta \hat{A}, \Delta \hat{B}\} \rangle_\alpha|^2$$

$$\geq \frac{1}{4} |\langle [\Delta \hat{A}, \Delta \hat{B}] \rangle_\alpha|^2$$

## § 2.3 量子化条件

正则量子化:

$$[q_\alpha, q_\beta] = 0, [p_\alpha, p_\beta] = 0, [q_\alpha, p_\beta] = i\hbar \delta_{\alpha\beta}$$

$$\hookrightarrow \Delta x \Delta p \geq \frac{1}{2} \hbar$$

$$[\hat{A}, \hat{B} + \hat{C}] = [\hat{A}, \hat{B}] + [\hat{A}, \hat{C}]$$

$$\begin{aligned} [\hat{A}, \hat{B}\hat{C}] &= \hat{A}\hat{B}\hat{C} - \hat{B}\hat{C}\hat{A} = \hat{A}\hat{B}\hat{C} - \hat{B}\hat{A}\hat{C} + \hat{B}\hat{A}\hat{C} - \hat{B}\hat{C}\hat{A} \\ &= [\hat{A}, \hat{B}]\hat{C} + \hat{B}[\hat{A}, \hat{C}] \end{aligned}$$

$$\vec{L} = \vec{r} \times \vec{p}$$

$$L_x = \hat{y}p_z - \hat{z}p_y, L_y = \hat{z}p_x - \hat{x}p_z, L_z = \hat{x}p_y - \hat{y}p_x$$

$$[L_x, L_y] = [\hat{y}p_z - \hat{z}p_x, \hat{z}p_x - \hat{x}p_z]$$

$$\Rightarrow [L_\alpha, L_\beta] = \epsilon_{\alpha\beta\gamma} i\hbar L_\gamma, \quad \vec{L} \times \vec{L} = i\hbar \vec{L}$$

## § 2.4 表象和表象变换

1. 选择可观测量的本征态  $|n\rangle$  为基矢

$$|\psi\rangle = \sum_n \psi_n |n\rangle = (|1\rangle \cdots |N\rangle) \begin{pmatrix} \psi_1 \\ \vdots \\ \psi_N \end{pmatrix}$$

$\{\psi_n\}$  称为  $|\psi\rangle$  在  $\hat{A}$  表象下的坐标。

表象变换

$$|\psi\rangle = \sum_n \psi_n |n\rangle = \sum_n \psi_n \langle n' | n \rangle |n'\rangle = \sum_{n'} \psi_{n'} |n'\rangle$$

$$(\psi_{n'}) = (\psi_n) \langle n' | n \rangle = U^\dagger (\psi_n)$$

变换矩阵:  $U = \langle n | n' \rangle$

$$|n'\rangle = \sum_n |n\rangle \langle n | n' \rangle$$

$$\hookrightarrow \{\psi_{n'}\} = \{\psi_n\} \hat{U} = \hat{U}^\dagger \{\psi_n\}$$

$$\hat{U} \hat{U}^\dagger = 1, \quad U \text{ 为么正矩阵.}$$

$$\begin{aligned} \text{证: } (\hat{U} \hat{U}^\dagger)_{ij} &= \sum_k U_{ik} U_{jk}^* = \sum_k U_{ik} U_{jk}^* \\ &= \sum_k \langle i | k \rangle \langle j | k \rangle^* = \sum_k \langle i | k \rangle \langle k | j \rangle \\ &= \delta_{ij} \end{aligned}$$

$$\text{左矢: } \langle \psi | = (\psi_1^* \cdots \psi_N^*) \begin{pmatrix} \langle 1 | \\ \vdots \\ \langle N | \end{pmatrix}$$

$$\begin{aligned} \text{内积: } \langle \psi | \phi \rangle &= \sum_n \langle \psi | n \rangle \langle n | \phi \rangle \\ &= (\psi_1^* \cdots \psi_N^*) \begin{pmatrix} \phi_1 \\ \vdots \\ \phi_N \end{pmatrix} \end{aligned}$$

力学量完全集:

一组彼此对易的力学量的集合

$$(\hat{A}_1, \hat{A}_2, \dots)$$

它们的共同本征态可唯一、完备地描述量子态。



## 2. 算符的表示

$$\hat{F}|\psi\rangle = |\phi\rangle$$

$$\phi_m = \langle m|\phi\rangle = \langle m|\hat{F}|\psi\rangle$$

$$= \sum_n \langle m|\hat{F}|n\rangle \langle n|\psi\rangle$$

$$= \sum_n F_{mn} \psi_n$$

$$\langle m'|\hat{F}|\hat{n}'\rangle = \sum_{m,n} \langle m'|m\rangle \langle m|\hat{F}|n\rangle \langle n|n'\rangle$$

$$= U_{mm'}^* F_{mn} U_{nn'}$$

$$\hookrightarrow \hat{F}' = U^\dagger \hat{F} U$$

若  $u$  为  $\hat{F}$  的本征矢量, 则  $\hat{F}'$  为  $\hat{F}$  经过么正变换后的对角化矩阵.

例: 2维 Hilbert 空间中有正交归一基矢  $|1\rangle, |2\rangle$

$$\text{算符: } \hat{A} = \frac{1}{2}(|1\rangle\langle 2| + |2\rangle\langle 1|), \quad \hat{B} = \frac{i}{2}(|1\rangle\langle 2| - |2\rangle\langle 1|)$$

(1) 证明:  $\hat{A}, \hat{B}$  是厄米算符.

(2) 求出  $\hat{B}$  表象中  $\hat{A}$  的表示

在表象  $|1\rangle, |2\rangle$  下,

$$\hat{A} = \begin{pmatrix} \langle 1|\hat{A}|1\rangle & \langle 1|\hat{A}|2\rangle \\ \langle 2|\hat{A}|1\rangle & \langle 2|\hat{A}|2\rangle \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\hat{B} = \begin{pmatrix} \langle 1|\hat{B}|1\rangle & \langle 1|\hat{B}|2\rangle \\ \langle 2|\hat{B}|1\rangle & \langle 2|\hat{B}|2\rangle \end{pmatrix} = \frac{i}{2} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

求出  $\hat{B}$  的本征矢量:

$$|\hat{B} - \lambda I| = \begin{vmatrix} -\lambda & \frac{i}{2} \\ -\frac{i}{2} & -\lambda \end{vmatrix} = \lambda^2 - \frac{1}{4} = 0 \rightarrow \lambda = \pm \frac{1}{2}$$

$$\begin{pmatrix} -\frac{1}{2} & \frac{i}{2} \\ -\frac{i}{2} & -\frac{1}{2} \end{pmatrix} \rightarrow \begin{pmatrix} -\frac{1}{2} & \frac{i}{2} \\ 0 & 0 \end{pmatrix} \text{ ①} \times i - \text{②} \rightarrow \frac{1}{2} \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} \frac{1}{2} & \frac{i}{2} \\ -\frac{i}{2} & \frac{1}{2} \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{2} & \frac{i}{2} \\ 0 & 0 \end{pmatrix} \text{ ①} \times i + \text{②} \rightarrow \frac{1}{2} \begin{pmatrix} 2 \\ -1 \end{pmatrix}$$

$$\lambda_1 = \frac{1}{2}, |\lambda_1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad \lambda_2 = -\frac{1}{2}, |\lambda_2\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 2 \\ -1 \end{pmatrix}$$

$$\hat{A}' = \begin{pmatrix} \langle \lambda_1|\hat{A}|\lambda_1\rangle & \langle \lambda_1|\hat{A}|\lambda_2\rangle \\ \langle \lambda_2|\hat{A}|\lambda_1\rangle & \langle \lambda_2|\hat{A}|\lambda_2\rangle \end{pmatrix} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$$

### 3. 坐标和动量表象

(1) 坐标算符  $\hat{x}$  的本征态:  $|x\rangle$ ,  $\hat{x}|x\rangle = x|x\rangle$

$\hat{x}|x'\rangle = x'|x'\rangle$   
 本征态  $|x'\rangle$  在表象  $|x\rangle$  下:  $\langle x|x'\rangle = \delta(x-x')$  正交归一的  $\{|x\rangle\}$  基

$\hat{x}$  的表示:  $\langle x|\hat{x}|x'\rangle = x'\langle x|x'\rangle = x\delta(x-x)$   $\sim x_i x_j = \delta_{ij}$

完备性条件:  $\int dx |x\rangle \langle x| = 1$

$\hookrightarrow |y\rangle = \int dx |x\rangle \langle x|y\rangle$ ,

$$\psi(x) = \langle x|y\rangle$$

(2) 动量算符的表示

引入辅助线性算符  $\hat{a}$  满足  $\langle \bar{x}|\hat{a}|y\rangle = \frac{\partial}{\partial x} \psi(x)$

$$\langle x|\hat{a}\hat{x}|y\rangle = \frac{\partial}{\partial x} \langle x|\hat{x}|y\rangle = \frac{\partial}{\partial x} [x \langle x|y\rangle]$$

$$= x \frac{\partial}{\partial x} \langle x|y\rangle + \langle x|y\rangle$$

$$= \langle x|\hat{x}\hat{a}|y\rangle + \langle x|y\rangle$$

$$\hookrightarrow \hat{a}\hat{x} = \hat{x}\hat{a} + 1, [\hat{x}, \hat{a}] = -1 \quad [\hat{x}, \hat{p}_x] = i\hbar$$

$$\hat{p}_x = -i\hbar \hat{a}$$

$$\langle \phi|\hat{a}|y\rangle = \int dx \langle \phi|x\rangle \langle x|\hat{a}|y\rangle$$

$$= \int dx \langle \phi|x\rangle \frac{\partial}{\partial x} \langle x|y\rangle$$

$$= \langle \phi|x\rangle \langle x|y\rangle \Big|_{-\infty}^{\infty} - \int \langle x|y\rangle \frac{\partial}{\partial x} \langle \phi|x\rangle$$

$$= - \int \langle x|y\rangle \frac{\partial}{\partial x} \langle \phi|x\rangle$$

$$= - \int dx \langle \phi|x\rangle^* \left( \frac{\partial}{\partial x} \langle x|y\rangle \right)^* = - \int dx \langle \phi|x\rangle^* \langle x|\hat{a}|y\rangle^*$$

$$= - \int dx \langle \phi|x\rangle \langle x|\hat{a}|y\rangle^*$$

$$= - \langle \phi|\hat{a}|y\rangle^* = - \langle \phi|\hat{a}^\dagger|y\rangle$$

$$\Rightarrow \hat{a}^\dagger = -\hat{a}, \quad \hat{p}_x^\dagger = \hat{p}_x$$

$\hat{p}_x$  在坐标表象下的表示:

$$\langle x|\hat{p}_x|x'\rangle = -i\hbar \langle x|\hat{a}|x'\rangle = -i\hbar \frac{\partial}{\partial x} \delta(x-x')$$

在  $|x\rangle$  表象下:

$$\begin{aligned}\langle x | \hat{p}_x | p_x \rangle &= \langle x | p_x | p_x \rangle = p_x \langle x | p_x \rangle \\ &= \int \langle x | \hat{p}_x | x' \rangle \langle x' | p_x \rangle dx'\end{aligned}$$

$|p_x\rangle$  为  $\hat{p}_x$  的本征态.

$$\Rightarrow -i\hbar \int \frac{\partial}{\partial x} \delta(x-x') \langle x' | p_x \rangle dx' = p_x \langle x | p_x \rangle$$

$$-i\hbar \frac{\partial}{\partial x} \langle x | p_x \rangle = p_x \langle x | p_x \rangle$$

$$\text{解得 } p_x(x) = \langle x | p_x \rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{\frac{i p_x x}{\hbar}} \quad \text{傅里叶变换.}$$

$$\text{动量表象下: } \langle p_x | p_x' \rangle = \delta(p_x - p_x')$$

$$\langle p_x | \hat{p}_x | p_x' \rangle = p_x \delta(p_x - p_x')$$

$\hat{x}$  在动量表象的表示:

$$\begin{aligned}\langle p_x | \hat{x} | p_x' \rangle &= \int dx dx' \langle p_x | x \rangle \langle x | \hat{x} | x' \rangle \langle x' | p_x' \rangle \quad I = \int dx |x\rangle \langle x| = \int dx' |x'\rangle \langle x'| \\ &= \int dx dx' \langle p_x | x \rangle x \delta(x-x') \langle x' | p_x' \rangle\end{aligned}$$

$$= \frac{1}{\sqrt{2\pi\hbar}} \int dx dx' x \delta(x-x') e^{-\frac{i p_x x}{\hbar}} e^{\frac{i p_x' x}{\hbar}}$$

$$= \frac{1}{\sqrt{2\pi\hbar}} \int dx x e^{\frac{i x}{\hbar} (p_x' - p_x)}$$

$$= \frac{i}{\sqrt{2\pi\hbar}} \int dx \frac{\partial}{\partial p_x} e^{\frac{i x}{\hbar} (p_x' - p_x)} = i\hbar \frac{\partial}{\partial p_x} \delta(p_x' - p_x)$$

$$\int_{-\infty}^{\infty} dx e^{\frac{i x}{\hbar} (p_x' - p_x)} = 2\pi \delta\left(\frac{p_x' - p_x}{\hbar}\right) = 2\pi\hbar \delta(p_x' - p_x)$$

(1) 坐标算符在动量表象  $|p_x\rangle$  下的表示为  $i\hbar \frac{\partial}{\partial p_x}$

态变量在  $\hat{x}$  和  $\hat{p}$  之间的变换 (傅里叶变换)

$$\psi(x) = \langle x | \psi \rangle = \langle x | p_x \rangle \langle p_x | \psi \rangle$$

$$= \frac{1}{\sqrt{2\pi\hbar}} \int dp_x e^{\frac{i p_x x}{\hbar}} \psi(p_x)$$

$$\psi(p_x) = \langle p_x | \psi \rangle = \langle p_x | x \rangle \langle x | \psi \rangle$$

$$= \frac{1}{\sqrt{2\pi\hbar}} \int dx e^{-\frac{i p_x x}{\hbar}} \psi(x)$$

$$\Rightarrow \text{类似: } \langle \vec{r} | \vec{r}' \rangle = \delta(\vec{r} - \vec{r}')$$

$$\langle \vec{p} | \vec{p}' \rangle = \delta(\vec{p} - \vec{p}')$$

$$\langle \vec{r} | \vec{r} | \vec{r}' \rangle = \vec{r} \delta(\vec{r} - \vec{r}')$$

$$\langle \vec{p} | \hat{p} | \vec{p}' \rangle = \vec{p} \delta(\vec{p} - \vec{p}')$$

$$\langle \vec{r} | \hat{p} | \vec{r}' \rangle = -i\hbar \nabla \delta(\vec{r} - \vec{r}')$$

$$\langle \vec{p} | \vec{r} | \vec{p}' \rangle = i\hbar \nabla \delta(\vec{p} - \vec{p}')$$

$$\langle \vec{r} | \vec{p} \rangle = \frac{1}{(2\pi\hbar)^{3/2}} e^{\frac{i \vec{p} \cdot \vec{r}}{\hbar}}$$

$$\langle \vec{p} | \vec{r} \rangle = \frac{1}{(2\pi\hbar)^{3/2}} e^{-\frac{i \vec{p} \cdot \vec{r}}{\hbar}}$$

## § 2.5 运动方程

1. 时间演化算符  $\hat{U}(t, t')$

$$|\psi(t')\rangle = \hat{U}(t', t) |\psi(t)\rangle$$

1)  $\hat{U}$  只与系统有关, 与具体的态无关

$$|\alpha(t)\rangle = c_1 |\alpha(t')\rangle + c_2 |\beta(t')\rangle$$

$$\hat{U} |\alpha(t)\rangle = c_1 \hat{U} |\alpha(t')\rangle + c_2 \hat{U} |\beta(t')\rangle$$

叠加原理  $\rightarrow \hat{U}$  是线性算符.

2)  $\hat{U}$  是么正算符:

$$\text{归一化条件 } \langle \psi(t) | \hat{U}^\dagger(t', t) \hat{U}(t, t') | \psi(t) \rangle$$

$$= \langle \psi(t) | \psi(t') \rangle = 1$$

$$\hookrightarrow \hat{U}(t, t') \hat{U}^\dagger(t', t) = 1$$

2. Schrodinger 运动方程.

$$\hat{U}(t + \delta t, t) = 1 + \delta t \Omega(t)$$

$$1 = \hat{U}^\dagger(t + \delta t, t) \hat{U}(t + \delta t, t)$$

$$= [1 + \delta t \Omega^\dagger(t)] [1 + \delta t \Omega(t)]$$

$$\approx 1 + \delta t [\Omega(t) + \Omega^\dagger(t)] \rightarrow \Omega(t) \text{ 是反厄米算符.}$$

$$\frac{\partial |\psi(t)\rangle}{\partial t} = \lim_{\delta t \rightarrow 0} \frac{|\psi(t + \delta t)\rangle - |\psi(t)\rangle}{\delta t}$$

$$= \lim_{\delta t \rightarrow 0} \frac{\hat{U}(t + \delta t, t) - 1}{\delta t} = \Omega(t)$$

$$\hat{H}(t) = i\hbar \Omega(t)$$

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H}(t) |\psi(t)\rangle$$

在给定表象下:

$$i\hbar \frac{\partial}{\partial t} \psi(\vec{r}, t) = \int d^3 r' \langle \vec{r} | \hat{H} | \vec{r}' \rangle \psi(\vec{r}', t)$$

坐标表象中势是局域的  $\langle \vec{r} | \hat{H} | \vec{r}' \rangle = H(\vec{r}, t) \delta(\vec{r} - \vec{r}')$  ? A:  $\hat{H}$  只有“对角元”

$\Rightarrow$  Schrodinger 方程:  $i\hbar \frac{\partial}{\partial t} \psi(\vec{r}, t) = H(\vec{r}, t) \psi(\vec{r}, t)$

没有长程相互作用, 那局域~

3. 时间演化算符的具体形式.

$$\psi(t) = \hat{U}(t, t_0) \psi(t_0)$$

$$\hookrightarrow \text{当 } \hat{H} \text{ 为 } \hat{U}(t, t_0) = \hat{T} \hat{U}(t, t_0)$$

$$\hat{U}(t, t_0) = 1 - \frac{i}{\hbar} \int_{t_0}^t dt \hat{H} \hat{U}(t, t_0)$$

迭代法:

$$\hat{U}(t, t_0) = \sum_{n=0}^{\infty} \left(-\frac{i}{\hbar}\right)^n \int_{t_0}^t dt_1 \int_{t_1}^{t_0} dt_2 \dots \int_{t_1}^{t_{n-1}} dt_n \hat{H} \hat{U}(t_n, t_0)$$

引入编时算符: 不同时刻  $\hat{H}$  不一定对易

$$\hat{T}[\hat{H}(t_1) \dots \hat{H}(t_n)] = \begin{cases} \hat{H}(t_1) \dots \hat{H}(t_n) & t_1 > \dots > t_n \\ \hat{H}(t_n) \dots \hat{H}(t_1) & t_n > \dots > t_1 \end{cases}$$

将积分变为:

$$\begin{aligned} \hat{U}(t, t_0) &= \sum_{n=0}^{\infty} \frac{1}{n!} \left(-\frac{i}{\hbar}\right)^n \int_{t_0}^t dt_1 \int_{t_1}^{t_0} dt_2 \dots \int_{t_1}^{t_{n-1}} dt_n \hat{T}[\hat{H}(t_1) \dots \hat{H}(t_n)] \\ &= \hat{T} \left[ e^{-\frac{i}{\hbar} \int_{t_0}^t \hat{H}(t') dt'} \right] \end{aligned}$$

不含时:

$$\hat{U}(t, t_0) = e^{-\frac{i}{\hbar} \hat{H}(t-t_0)}$$

Hamilton 量本征态:

$$|\phi_n(t)\rangle = e^{-\frac{i}{\hbar} E_n(t-t_0)} |\phi_n(t_0)\rangle$$

$\Rightarrow$  研究态矢量随时间的演化, 可以在能量表象展开.

$$|\psi(t_0)\rangle = \sum_n c_n |\phi_n(t_0)\rangle$$

$$|\psi(t)\rangle = \sum_n c_n |\phi_n(t)\rangle$$

例. 考虑 Hilbert 空间和系统, 基矢为  $\{|1\rangle, |2\rangle, |3\rangle\}$

$$\hat{H} = \hbar\omega \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} \quad \hat{A} = a \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \quad \hat{B} = b \begin{pmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

$t=0$  时刻系统状态为  $|\psi(0)\rangle = \frac{1}{\sqrt{2}}|1\rangle + \frac{1}{2}|2\rangle + \frac{1}{2}|3\rangle$

(1) 计算  $t=0$  时刻  $\Delta H = \sqrt{\langle \hat{H}^2 \rangle - \langle \hat{H} \rangle^2}$

$$\hat{H}^2 = \hbar^2\omega^2 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{pmatrix}$$

$$\begin{aligned} \langle \hat{H}^2 \rangle &= \langle \psi | \hat{H}^2 | \psi \rangle = \hbar^2\omega^2 \left( \frac{1}{\sqrt{2}} \frac{1}{2} \frac{1}{2} \right) \begin{pmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix} \\ &= \hbar^2\omega^2 \left( \frac{1}{2} \ 2 \ 2 \right) \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix} = \frac{5}{2} \hbar^2\omega^2 \end{aligned}$$

$$\begin{aligned} \langle \hat{H} \rangle &= \langle \psi | \hat{H} | \psi \rangle = \hbar\omega \left( \frac{1}{\sqrt{2}} \frac{1}{2} \frac{1}{2} \right) \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix} \\ &= \hbar\omega \left( \frac{1}{\sqrt{2}} \ 1 \ 1 \right) \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix} = \frac{3}{2} \hbar\omega \end{aligned}$$

$$\Delta H = \sqrt{\langle \hat{H}^2 \rangle - \langle \hat{H} \rangle^2} = \frac{1}{2} \hbar\omega$$

$$(2) |\mathbf{A} - \lambda a \mathbf{I}| = a^3 \begin{vmatrix} 1-\lambda & 0 & 0 \\ 0 & -\lambda & 1 \\ 0 & 1 & -\lambda \end{vmatrix} = a^3 [(1-\lambda)(\lambda^2-1)] = 0$$

$$\hookrightarrow \lambda = 1 \text{ 或 } \lambda = -1$$

$$\lambda = 1 \text{ 时, } \begin{pmatrix} 0 & 0 & 0 \\ 0 & -1 & 1 \\ 0 & 1 & -1 \end{pmatrix} \rightarrow \begin{pmatrix} 0 & -1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad |\alpha_1\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad |\alpha_2\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

$$\lambda = -1 \text{ 时, } \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \quad |\alpha_3\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$$

$$\alpha_1 = \langle \alpha_1 | \psi(0) \rangle^2 + \langle \alpha_2 | \psi(0) \rangle^2 = \frac{1}{2} + \frac{1}{2} \left( \frac{1}{2} + \frac{1}{2} \right) = 1$$

$$\alpha_2 = \langle \alpha_3 | \psi(0) \rangle^2 = 0$$

$$(3) \hat{A}\hat{B} = ab \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} = ab \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

$$\hat{B}\hat{A} = ba \begin{pmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} = ab \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

$$\begin{aligned} |\mathbf{B} - \lambda b \mathbf{I}| &= b \begin{vmatrix} -\lambda & 1 & 1 \\ 1 & 1-\lambda & 0 \\ 1 & 0 & 1-\lambda \end{vmatrix} = -\lambda(1-\lambda)^2 - (1-\lambda) - (1-\lambda) = [-\lambda(1-\lambda) - 2](1-\lambda) \\ &= (\lambda^2 - \lambda - 2)(1-\lambda) = (\lambda+1)(\lambda-2)(1-\lambda) \end{aligned}$$

$$\lambda = -1 \quad \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 0 \\ 1 & 0 & 2 \end{pmatrix} \rightarrow \begin{pmatrix} 0 & 0 & 0 \\ 1 & 2 & 0 \\ 1 & 0 & 2 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 2 & 0 \\ 0 & -2 & 2 \\ 0 & 0 & 0 \end{pmatrix} \quad |\beta_1\rangle = \frac{1}{\sqrt{6}} \begin{pmatrix} 2 \\ -1 \\ -1 \end{pmatrix}$$

$$\lambda = 2 \quad \begin{pmatrix} -2 & 1 & 1 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix} \quad |\beta_2\rangle = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\lambda = 1 \quad \begin{pmatrix} -1 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \quad |\beta_3\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

共同本征态  $|\beta_3\rangle \sim |\alpha_3\rangle$ , 继续求出剩下两个

在  $A$  的本征子空间中:

$$\textcircled{1} \lambda = 1$$

$$|\alpha_1\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad |\alpha_2\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

$$B \text{ 的限制: } B' = \begin{pmatrix} \langle \alpha_1 | B | \alpha_1 \rangle & \langle \alpha_1 | B | \alpha_2 \rangle \\ \langle \alpha_2 | B | \alpha_1 \rangle & \langle \alpha_2 | B | \alpha_2 \rangle \end{pmatrix} = \begin{pmatrix} 0 & \sqrt{2} \\ \sqrt{2} & 1 \end{pmatrix}$$

$$|B' - \lambda I| = \begin{vmatrix} -\lambda & \sqrt{2} \\ \sqrt{2} & 1-\lambda \end{vmatrix} = \lambda(\lambda-1) - 2 = (\lambda-2)(\lambda+1)$$

$$\lambda_1 = 2 \quad \begin{pmatrix} -2 & \sqrt{2} \\ \sqrt{2} & -1 \end{pmatrix} \rightarrow |\beta_1'\rangle = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ \sqrt{2} \end{pmatrix}$$

$$\lambda_2 = -1 \quad \begin{pmatrix} 1 & \sqrt{2} \\ \sqrt{2} & 2 \end{pmatrix} \rightarrow |\beta_2'\rangle = \frac{1}{\sqrt{3}} \begin{pmatrix} \sqrt{2} \\ -1 \\ 1 \end{pmatrix}$$

$$\Rightarrow A, B \text{ 的共同本征态: } \frac{1}{\sqrt{3}} |\alpha_1\rangle + \frac{\sqrt{2}}{3} |\alpha_2\rangle \text{ 和 } \frac{\sqrt{2}}{3} |\alpha_1\rangle - \frac{1}{\sqrt{3}} |\alpha_2\rangle$$

$$\textcircled{2} \lambda = -1 \quad |\alpha_3\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$$

$$B'' = \langle \alpha_3 | B | \alpha_3 \rangle = \frac{1}{2} (0 \ 1 \ -1) \begin{pmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} = \frac{1}{2} (0 \ 1 \ -1) \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} = 1$$

$$\Rightarrow A, B \text{ 的共同本征态: } |\alpha_3\rangle$$

$$(4) |\delta_1\rangle = \frac{1}{\sqrt{3}} |\alpha_1\rangle + \frac{\sqrt{2}}{3} |\alpha_2\rangle = \frac{\sqrt{3}}{3} \begin{pmatrix} 1 \\ 1 \\ \sqrt{2} \end{pmatrix}$$

$$|\delta_2\rangle = \frac{\sqrt{2}}{3} |\alpha_1\rangle - \frac{1}{\sqrt{3}} |\alpha_2\rangle = \frac{\sqrt{3}}{3} \begin{pmatrix} \sqrt{2} \\ -1 \\ 1 \end{pmatrix} = \frac{\sqrt{6}}{6} \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$$

$$|\delta_3\rangle = |\alpha_3\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$$

$$U = (|\delta_1\rangle, |\delta_2\rangle, |\delta_3\rangle) = \begin{pmatrix} \frac{\sqrt{3}}{3} & \frac{\sqrt{6}}{6} & 0 \\ \frac{\sqrt{3}}{3} & -\frac{\sqrt{6}}{6} & \frac{1}{\sqrt{2}} \\ \frac{\sqrt{3}}{3} & \frac{\sqrt{6}}{6} & -\frac{1}{\sqrt{2}} \end{pmatrix}$$

$$\tilde{H}' = U^\dagger \tilde{H} U$$

$$= \begin{pmatrix} \frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{3} \\ \frac{\sqrt{6}}{3} & -\frac{\sqrt{6}}{6} & -\frac{\sqrt{6}}{6} \\ 0 & \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix} \hbar \omega \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} \frac{\sqrt{3}}{3} & \frac{\sqrt{6}}{6} & 0 \\ \frac{\sqrt{3}}{3} & -\frac{\sqrt{6}}{6} & \frac{1}{\sqrt{2}} \\ \frac{\sqrt{3}}{3} & \frac{\sqrt{6}}{6} & -\frac{1}{\sqrt{2}} \end{pmatrix}$$

$$= \hbar \omega \begin{pmatrix} \frac{\sqrt{3}}{3} & \frac{2\sqrt{3}}{3} & \frac{2\sqrt{3}}{3} \\ \frac{\sqrt{6}}{6} & -\frac{\sqrt{6}}{3} & \frac{\sqrt{6}}{3} \\ 0 & \sqrt{2} & -\sqrt{2} \end{pmatrix} \begin{pmatrix} \frac{\sqrt{3}}{3} & \frac{\sqrt{6}}{6} & 0 \\ \frac{\sqrt{3}}{3} & -\frac{\sqrt{6}}{6} & \frac{1}{\sqrt{2}} \\ \frac{\sqrt{3}}{3} & \frac{\sqrt{6}}{6} & -\frac{1}{\sqrt{2}} \end{pmatrix}$$

$$= \hbar \omega \begin{pmatrix} \frac{1}{3} & -\frac{1}{3} & 0 \\ -\frac{\sqrt{2}}{3} & -2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

$$(5) \quad |a_1\rangle = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \quad |a_2\rangle = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \quad |a_3\rangle = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

$$|\psi(0)\rangle = \langle a_1 | \psi(0) \rangle |a_1\rangle + \dots$$

$$= 1 |a_1\rangle + \left(\frac{1}{\sqrt{2}} + \frac{1}{2}\right) |a_2\rangle + \left(\frac{1}{\sqrt{2}} + \frac{1}{2}\right) |a_3\rangle$$

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}} |1\rangle e^{-i\omega t} + \frac{1}{2} |2\rangle e^{-i2\omega t} + \frac{1}{2} |3\rangle e^{-i\omega t}$$

$$\langle \bar{B}(t) \rangle = \langle \psi(t) | \bar{B}(t) | \psi(t) \rangle$$

$$= \left( \frac{1}{\sqrt{2}} e^{i\omega t} \quad \frac{1}{2} e^{i2\omega t} \quad \frac{1}{2} e^{i\omega t} \right) b \begin{pmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} e^{-i\omega t} \\ \frac{1}{2} e^{-i2\omega t} \\ \frac{1}{2} e^{-i\omega t} \end{pmatrix}$$

$$= b \left( e^{i\omega t} \frac{1}{\sqrt{2}} e^{i\omega t} + \frac{1}{2} e^{i2\omega t} \frac{1}{\sqrt{2}} e^{i\omega t} + \frac{1}{2} e^{i\omega t} \frac{1}{2} e^{i\omega t} \right) \begin{pmatrix} \frac{1}{\sqrt{2}} e^{-i\omega t} \\ \frac{1}{2} e^{-i2\omega t} \\ \frac{1}{2} e^{-i\omega t} \end{pmatrix}$$

$$= b \left[ \frac{1}{\sqrt{2}} e^{i\omega t} + 2 \left( \frac{1}{2\sqrt{2}} e^{-i\omega t} + \frac{1}{4} \right) \right]$$

$$= b \left( \frac{1}{2} + \sqrt{2} \cos \omega t \right)$$